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From Pressure Sink to Production Sync: A Strategic Well Pattern Optimization (WPO) of a Mature Onshore Oil Field in the UAE for Restoring Reservoir Health, Maximizing Facilities Utilization and Embracing Digital Transformation

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Abstract

Field development reviews frequently uncover inefficiencies in mature reservoirs due to incomplete well patterns, poor sweep efficiency, and uneven pressure support. In Reservoir-X, these challenges were impacting recovery performance and long-term development planning. The objective of this initiative was to proactively address these inefficiencies through a Well Pattern Optimization (WPO) framework. The scope involved identifying pressure sink areas and regions of low sweep efficiency, optimizing injector-producer alignments, and integrating digital technologies to enhance surveillance and maintain reservoir health.

The Well Pattern Optimization (WPO) strategy began with detailed reservoir monitoring and simulation analysis to diagnose pressure-depleted zones and potential poorly swept areas. An adaptive injector-producer optimization plan was developed to improve areal and vertical sweep efficiency utilizing the current slots and surface facilities. As part of the digitalization, advanced dashboards were deployed to review the current patterns, detect performance deviations, and recommend remedial actions. Additionally, Autonomous Control System (ACS) "an AI-based automation system" was implemented for the first time in a gas-lifted reservoir to enable wells to self-adjust gas-lift choke settings, ensuring operations remain within defined envelopes and dynamically adapt to field conditions.

The optimized well pattern configuration resulted in notable improvements in sweeping efficiency and reservoir pressure management. By refining the placement and spacing of wells, the new configuration minimized irregular fluid movement and improved the overall displacement of oil, reducing bypassed oil volumes. Simulation results validated these enhancements, showing better pattern conformance and more uniform pressure support across the reservoir. Additionally, the implementation of digital dashboards significantly boosted operational visibility. These dashboards allowed to monitor real-time performance, enabling quicker diagnostics and proactive interventions, which further optimized field operations. The

deployment of ACS AI technology played a crucial role by automating well behavior, particularly enhancing gas-lift operations. This automation helped maintain stable reservoir energy, leading to an incremental production gain of 1-2% and a 5% improvement in gas-lift efficiency. Collectively, these initiatives not only restored short-term production performance but also strengthened Reservoir-X's outlook for long-term development. With reduced CAPEX and OPEX requirements, the improvements are projected to add approximately 9 million stock tank barrels (MMSTB) of cumulative oil and extend the production plateau by more than 10 months.

The integrated approach combining pattern optimization, real-time surveillance, and AI-enabled control demonstrates a scalable solution for other reservoirs facing similar challenges. The ability to align subsurface engineering with digital transformation reflects a modern standard for mature field amendment and exemplifies how targeted innovation can unlock previously untapped recovery potential while enhancing operational resilience.

Introduction

Field Development Planning (FDP)

Field Development Planning (FDP) is a structured, multidisciplinary process that defines how an oil or gas field will be developed and produced over its economic life. It aims to optimize hydrocarbon recovery while balancing technical feasibility, economic performance, operational risks, and environmental sustainability. FDP is not a one-time task but an evolving framework that adapts to new data, changing reservoir behavior, and market dynamics (Kaiser and Pulsipher, 2004). The process typically spans several stages from concept selection to detailed engineering and requires the integration of subsurface characterization, well planning, production forecasting, facility design, project economics, and regulatory compliance.

The foundation of FDP lies in robust subsurface evaluation, which includes static modeling based on geological and petrophysical interpretation and dynamic modeling using reservoir simulation. These models help to define original hydrocarbons in place (OOIP/OGIP), drive estimates of recovery factors, and support evaluation of different development strategies (Begg et al., 2001). Several development concepts are usually considered at the early stages such as vertical vs. horizontal wells, waterflood vs. gas injection, or conventional vs. enhanced oil recovery (EOR) techniques. These concepts are assessed through scenario modeling and ranked based on key performance indicators such as Net Present Value (NPV), Internal Rate of Return (IRR), payout time, and recovery efficiency.

Well planning is a central component of FDP. Engineers evaluate trajectories, spacing, number of wells, and injector-producer alignment to optimize reservoir contact and drainage. In mature or complex reservoirs, well placement decisions are guided by advanced tools like 3D geocellular models, streamline simulation, and geosteering technologies (Jafarpour and He, 2009). In offshore and remote developments, the well plan must also account for drilling logistics, rig availability, and surface constraints. Increasingly, intelligent completions, multilateral wells, and extended-reach drilling are incorporated into FDPs to enhance reservoir access and reduce footprint.

Surface facility planning is integrated into the FDP process to ensure that infrastructure such as separators, pipelines, compressors, and injection systems is appropriately sized and phased to match production forecasts. Key decisions involve facility type (modular, centralized), fluid handling capacity, export routes, power requirements, and potential for gas reinjection or flaring minimization. The facility design must align with environmental regulations and allow for future field expansion or debottlenecking.

Another crucial aspect is economic evaluation and risk analysis. FDP incorporates detailed cost estimates, production profiles, and sensitivity analysis to assess project economics under different price scenarios, cost assumptions, and subsurface uncertainties (Begg et al., 2001). Decision trees, Monte Carlo simulations, and probabilistic forecasting are often employed to quantify risks and guide investment decisions. Value of Information (VoI) analysis may also be conducted to determine the worth of additional data acquisition (e.g., appraisal wells, seismic surveys) before committing to full-scale development.

Modern FDPs increasingly adopt digital and data-driven technologies to enhance planning quality and speed. These include integrated asset modeling, machine learning algorithms for scenario optimization, cloud-based collaboration platforms, and real-time data from smart wells and digital twins. Such technologies enable dynamic field development strategies that are continuously refined as more data becomes available during early production phases.

Field Development Planning is a critical, holistic process that shapes the technical, economic, and operational trajectory of a hydrocarbon asset. Its success depends on accurate reservoir characterization, integrated cross-functional collaboration, rigorous economic modeling, and adaptive decision-making enabled by modern technologies. A well-executed FDP not only improves recovery, and project returns but also mitigates risks and ensures alignment with long-term field objectives.

Fig. 1 illustrates standard field development planning workflows and economic analysis logic, based on industry practices (Kaiser and Pulsipher, 2004; Begg et al., 2001).

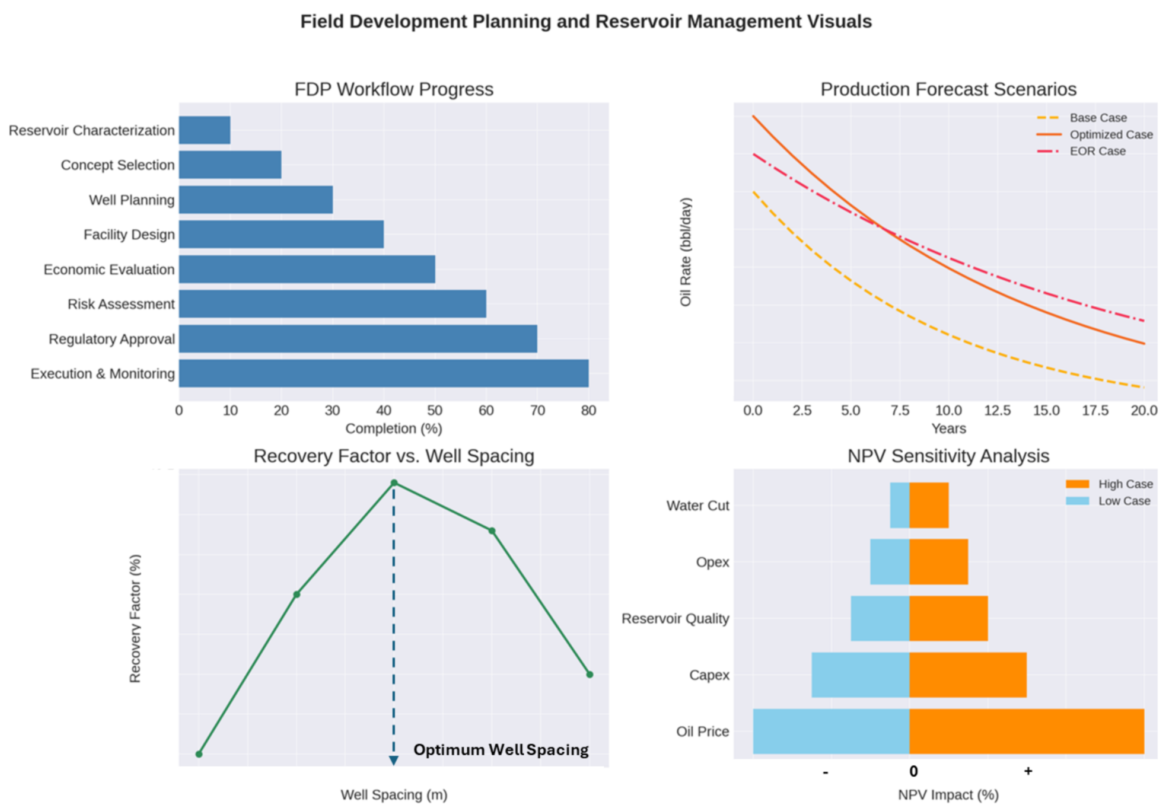

Figure 1—Standard Field Development Planning Workflows and Economic Analysis Logic

Four visualizations illustrating key aspects of Field Development Planning (FDP):

1. FDP Workflow Progress – Highlights the main planning stages from characterization to execution.
2. Production Forecast Scenarios – Compares output projections for different development strategies.
3. Recovery Factor vs. Well Spacing – Shows the impact of well spacing on reservoir recovery.
4. NPV Sensitivity Analysis – Displays how different parameters affect project economics.

Well Patterns (WP)

Well patterns are fundamental elements in the design and implementation of reservoir development strategies, particularly in waterfloods, gas floods, and enhanced oil recovery (EOR) projects. A well pattern refers to the spatial arrangement of injection and production wells, which is engineered to promote efficient fluid displacement and pressure maintenance across the reservoir. The primary goal is to optimize areal sweep efficiency, ensuring uniform fluid movement and minimizing bypassed oil. Traditional patterns include five-spot, seven-spot, nine-spot, and line-drive configurations. In a five-spot pattern, for example, one injection well is surrounded by four producers placed at the corners of a square, enabling radial displacement of fluids. Meanwhile, line-drive patterns are better suited for long, narrow reservoirs or when reservoir heterogeneity necessitates linear sweep fronts (Lake et al., 2014).

Another variation is the seven-spot pattern, which consists of a central injector surrounded by six producers forming a hexagon. This layout offers more injection and production capacity compared to the five-spot, making it suitable for larger or higher-permeability reservoirs. For even greater control over fluid movement, the nine-spot pattern is employed. It features one central well surrounded by eight others in a square grid, usually arranged with one injector and eight producers or vice versa. This pattern provides enhanced displacement efficiency and improves areal coverage, although it involves higher drilling and operational costs.

The line drive pattern is another commonly used approach, especially effective in reservoirs with directional permeability or a noticeable structural dip. In this setup, wells are aligned in straight parallel rows, with injectors and producers placed in alternating lines. The direct line drive positions injectors updip and producers downdip, leveraging gravity to aid fluid movement, while the inverted version does the opposite. When there is a need to maintain reservoir pressure around the perimeter, the peripheral pattern (or edge waterflood) is applied. In this pattern, injection wells are placed around the boundary of the reservoir to push oil toward centrally located producers. It is particularly effective in large, consolidated reservoirs.

In irregular or smaller reservoirs where traditional square-based patterns are not feasible, a triangular pattern may be used. This pattern involves placing wells in an equilateral triangle formation, typically with one injector and two producers, offering a flexible approach to coverage despite less uniform sweeping. Another useful variant is the staggered line drive pattern, which offsets the rows of injection and production wells rather than placing them directly opposite each other. This design improves contact with the reservoir in cases of complex geology, layering, or faulting.

Ultimately, the choice of a well pattern is influenced by various factors including reservoir heterogeneity, dip, fluid mobility, surface facility constraints, and economic considerations. A well-designed pattern enhances oil recovery, maintains reservoir pressure, and ensures more efficient utilization of injected fluids. Fig. 2 illustrates different types of wall patterns.

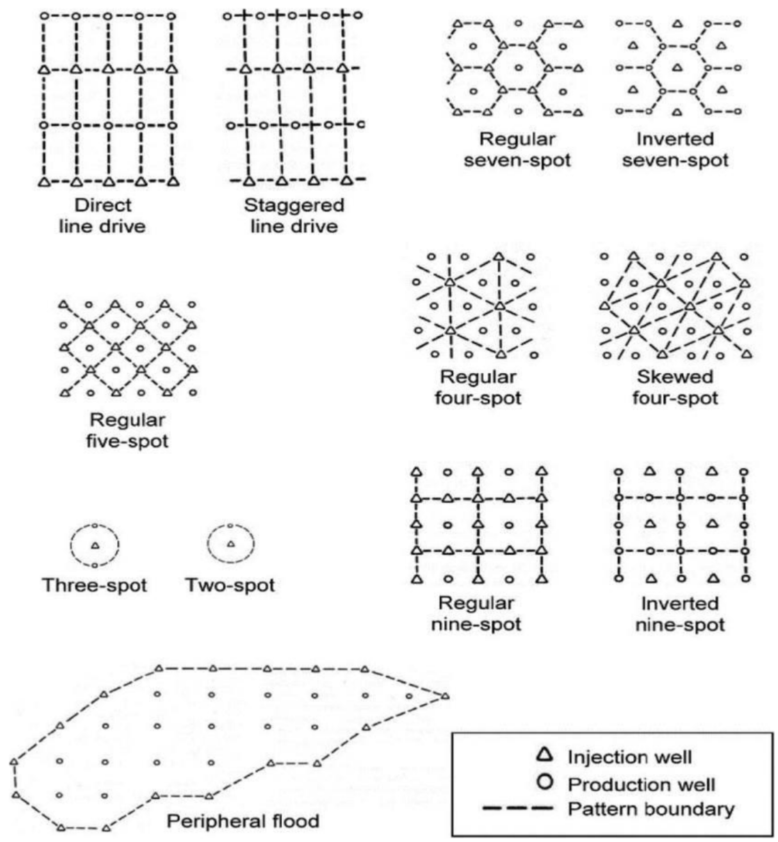

Figure 2—Different Types of Well Patterns (Lahkar, Nitin 2019)

Reservoir Management

Reservoir management is a multidisciplinary, continuous process that integrates technical expertise, data analysis, and operational decision-making to optimize hydrocarbon recovery and maximize economic value over the reservoir's life cycle. It combines elements of reservoir engineering, geology, geophysics, production operations, and data analytics to plan and manage reservoir development in a systematic and efficient manner (Lake et al., 2014). The goal is not only to enhance recovery but also to maintain reservoir pressure, control water and gas production, ensure conformance, and reduce operational risks.

A robust reservoir management strategy typically begins with a comprehensive static and dynamic characterization of the reservoir. This includes building detailed geological models, petrophysical evaluation, fluid property analysis, and history matching through dynamic simulation. Once a development plan is implemented, continuous monitoring becomes essential to track reservoir behavior and identify deviations from expected performance. Traditional surveillance tools include pressure transient analysis (PTA), rate transient analysis (RTA), and production logging tools (PLTs), which provide insights into fluid flow, reservoir heterogeneities, and well performance (Gao et al., 2005). In waterflood or gas injection projects, interwell tracer studies and water cut trends are used to assess sweep efficiency and identify channeling or thief zones.

4D seismic (time-lapse seismic) is another powerful monitoring technique, particularly in large-scale or offshore developments. It enables the visualization of fluid movement and pressure fronts over time, helping to refine simulation models and optimize injection and production strategies (Lumley, 2001). Reservoir simulation models remain a cornerstone of management, serving as predictive tools for scenario analysis, forecasting, and optimization.

The digital transformation of the oil and gas industry has further expanded the reservoir management toolkit. Digital oilfield technologies now allow for real-time monitoring of reservoir and well conditions using permanent downhole gauges, smart completions, and fiber optic sensing. These tools provide continuous data on pressure, temperature, flow rate, and other critical parameters, enabling engineers to make timely and informed decisions (Gomez et al., 2019). Machine learning and data analytics platforms are increasingly being used to detect production anomalies, predict reservoir behavior, and automate surveillance workflows (SPE, 2020).

Moreover, pattern-based surveillance approaches, especially in mature waterflooded fields, allow for the evaluation of injector-producer connectivity and pattern efficiency, leading to better conformance control and reservoir contact. These workflows integrate spatial production data, streamline simulation, tracer response, and pressure mapping to pinpoint underperforming patterns and guide targeted interventions (Alhuthali et al., 2007; Jafarpour et al., 2009).

Fig. 3 shows the typical reservoir management priorities observed in literature (Lake et al., 2014; Gomez et al., 2019).

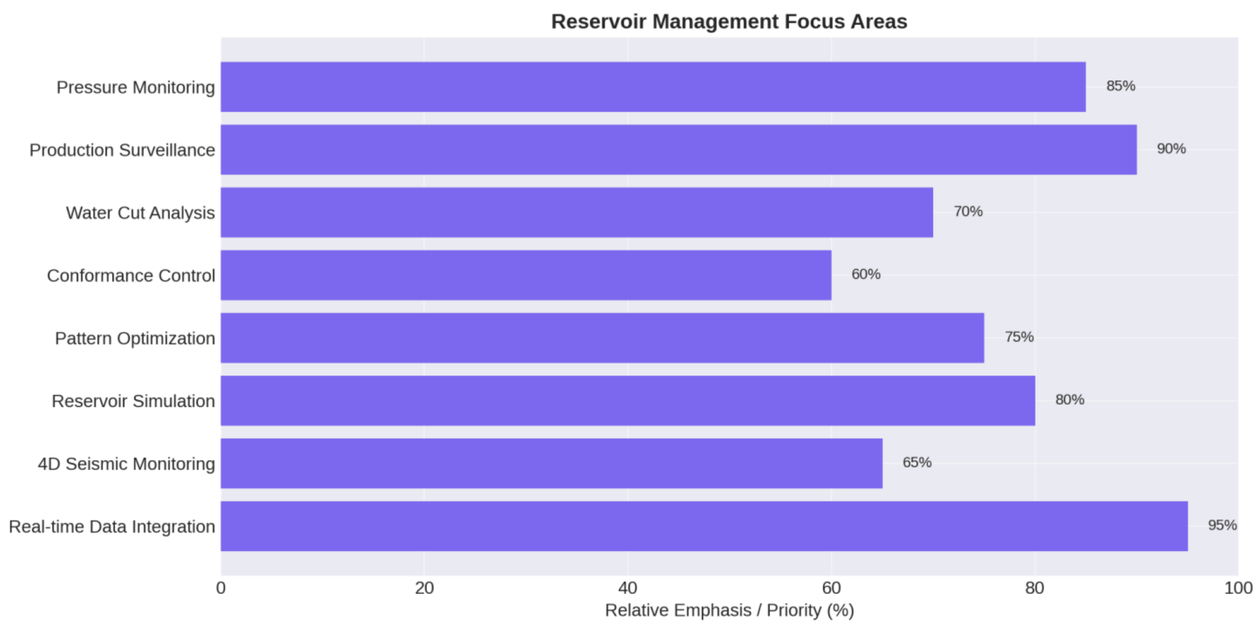

Figure 3—Typical Reservoir Management Priorities

Well Patterns Optimization (WPO)

While standard patterns serve as a baseline, they often require customization during the reservoir life cycle. Over time, due to changes in reservoir pressure distribution, fluid saturation, and operational practices, the efficiency of these patterns can degrade. Uneven injection profiles, preferential flow paths, and high-permeability streaks can lead to early water or gas breakthrough, poor displacement efficiency, and isolated pressure sink zones. These inefficiencies are particularly common in mature and heterogeneous reservoirs.

To address these challenges, Well Pattern Optimization (WPO) has become a critical component of modern reservoir management. WPO involves systematic evaluation and reconfiguration of well patterns to improve reservoir contact, enhance sweep, and balance injection and production rates. Optimization actions may include realigning injector-producer pairs, drilling infill or sidetrack wells, re-perforating or shutting off high water-cut intervals, and adjusting injection/production constraints based on real-time surveillance. The process is typically supported by data such as water cut trends, pressure maps, rate-transient diagnostics, streamline simulations, and tracer tests (Alhuthali et al., 2007). For example, pressure transient analysis can highlight areas of poor communication, while streamline simulation can visualize flow paths and reveal bypassed zones.

Modern WPO approaches increasingly leverage digital technologies to enhance decision-making. Automated pattern surveillance dashboards, machine learning algorithms, and AI-driven pattern ranking tools are used to rapidly evaluate large numbers of patterns and identify underperformers. These platforms integrate production data, spatial relationships, and historical interventions to prioritize optimization actions. Additionally, some operators have begun implementing dynamic WPO frameworks, which continuously adjust pattern configurations in response to real-time data and predictive modeling (Gomez et al., 2019).

Moreover, pattern performance is often linked to the concept of pattern balancing, which involves ensuring that the volume of fluids injected matches the withdrawal within a pattern, adjusted for compressibility and fluid expansion. Poor pattern balance can result in inefficient sweep, reservoir damage, or excessive water production. As such, pattern balancing calculations and voidage replacement ratio (VRR) monitoring are often integrated into WPO workflows. In summary, well pattern optimization is a dynamic, data-driven process that plays a central role in improving recovery efficiency, especially in secondary and tertiary recovery projects. By leveraging historical performance data, simulation models, and emerging digital technologies, WPO helps operators unlock bypassed oil, reduce water handling costs, and prolong the economic life of their assets.

Case Study of Reservoir-X Well Pattern Optimization

Field development evaluations and reviews often reveal significant inefficiencies in mature reservoirs, typically resulting from incomplete well patterns, uneven pressure support, and poor sweep efficiency (Ahmed and Meehan, 2016; Lake et al., 2014). These challenges contribute to suboptimal hydrocarbon recovery, bypassed oil zones, and complications in future development planning. In Reservoir-X, similar issues were identified, stemming from historical development constraints, reservoir heterogeneity, and limited optimization of injector-producer configurations. These inefficiencies are commonly observed in aging assets where legacy infrastructure and historical decisions continue to impact recovery performance (Al-Mutairi et al., 2017).

To address these challenges, a Well Pattern Optimization (WPO) initiative was launched in Reservoir-X in one of the onshore oil fields in the United Arab Emirates, aiming to enhance reservoir performance through targeted improvements in pattern alignment and reservoir management. The approach focused on diagnosing pressure-depleted areas and identifying zones with poor areal sweep, followed by strategic realignment of injectors and producers to improve connectivity and displacement efficiency.

Reservoir-X currently relies on two primary well patterns for development: Area-1 has five-spot configuration and Area-2 has line-drive arrangements. While these designs have provided a structural framework for production and injection, they have also resulted in uneven well spacing across various sectors of the reservoir. In some areas, the gaps between wells are too large, leaving zones inadequately supported by injectors and failing to maintain uniform pressure. This imbalance leads to the formation of low-pressure sink zones, where reservoir energy drops significantly, triggering operational challenges such as excessive gas liberation and gas coning, which drive up the Gas–Oil Ratio (GOR), as well as water breakthrough that contributes to a rising Water Cut (WCUT).

Beyond these production inefficiencies, the incomplete sweep leaves substantial volumes of oil trapped between injectors, causing high residual oil saturation and reducing ultimate recovery. Over time, this inefficiency not only threatens the ability to sustain the production plateau but also increases the risk of early field decline. To mitigate these issues, a comprehensive well pattern optimization strategy was essential to focus on improving injector–producer alignment, infilling underdeveloped zones, and balancing pressure distribution across the reservoir. Such an approach would strengthen pressure maintenance in identified sink areas, reduce GOR and WCUT risks, and tap into bypassed oil pockets, ultimately enhancing recovery efficiency and supporting long-term production targets even when injection rates fluctuate.

The Well Pattern Optimization (WPO) strategy for Reservoir-X was developed after a comprehensive analysis of the performance of each area, focusing on production trends, reservoir pressure behavior, sweep efficiency, and fluid movement dynamics. The evaluation highlighted that certain zones particularly in Blocks 1, 2, 3, and 4 were underperforming due to incomplete well patterns and long well spacing, which created pressure sinks and left large volumes of oil unswept (Fig.5). Detailed reservoir surveillance and simulation modeling were conducted to quantify the impact of these gaps and to test potential improvement scenarios. These studies confirmed that without intervention, production decline would accelerate, GOR and WCUT would continue to rise, and the reservoir's ability to sustain the production plateau would be compromised.

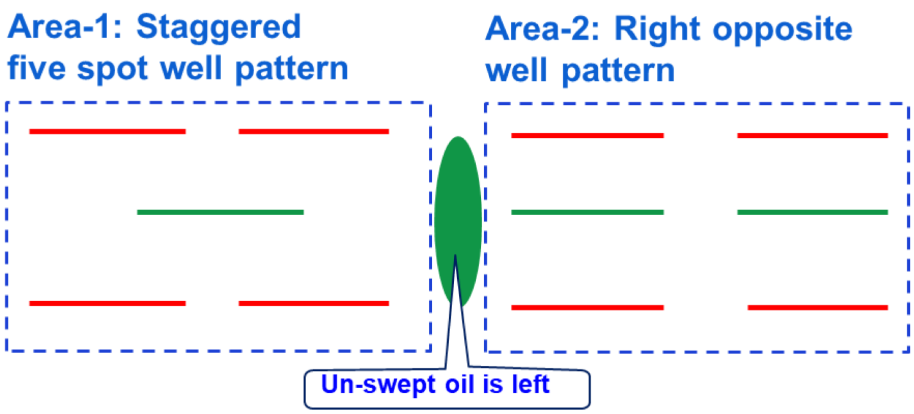

Figure 4—Well Patterns Transition Between Area-1 and Area-2

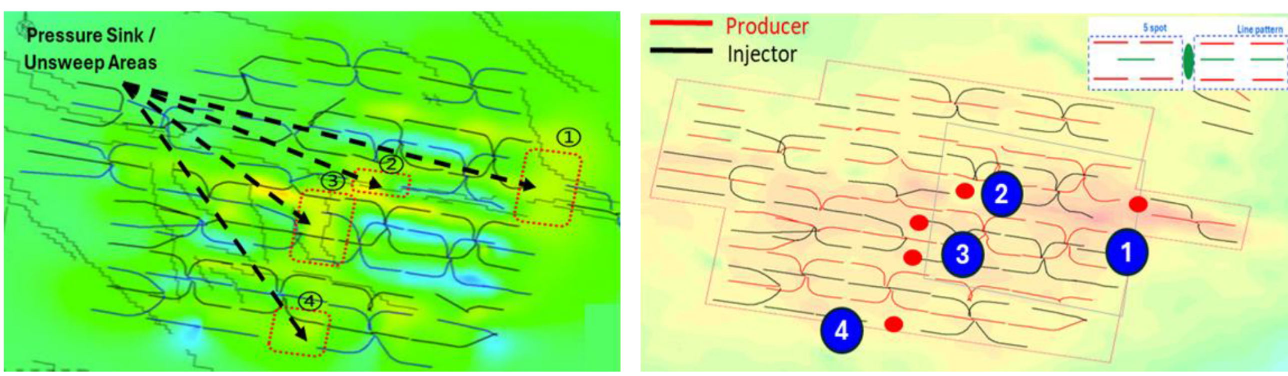

Figure 5—Identified Incomplete Patterns in Reservoir-X

To ensure cost efficiency, the pattern optimization strategy considers the current well slots, existing injectors, and available infrastructure, thereby minimizing additional CAPEX requirements. By leveraging existing facilities and aligning new wells with current development footprints, the approach reduces the need for extensive surface modifications or new facility installations. This not only accelerates project implementation but also maximizes the value of prior investments while enhancing recovery performance.

Building on these findings, the WPO strategy evolved into a targeted set of recommendations aimed at achieving the optimal development plan for the reservoir. The analysis supported extending the horizontal sections of existing injectors to improve reservoir contact and strengthen pressure support, as well as introducing Maximum Reservoir Contact (MRC) wells in strategically identified gaps to complete the existing well patterns. This dual approach ensures that under-swept areas are effectively drained, pressure distribution is balanced, and remaining oil pockets are accessed more efficiently.

Simulation results validated the proposed recommendations, projecting an incremental oil gain of approximately 9 million barrels and an extension of the production plateau by 10 months. More importantly, the simulations confirmed that reservoir pressure would be restored across these areas, mitigating the risks of high GOR and WCUT while optimizing waterflood performance. This area-by-area performance assessment, followed by the validation of each intervention through modeling, ensures that the WPO is not just a generic solution, but the optimum development recommendation that maximizes recovery, extends field life, and sustains production targets in a cost-effective and technically sound manner.

Fig. 6 shows the enhancement in the reservoir pressure after applying the WPO recommendations.

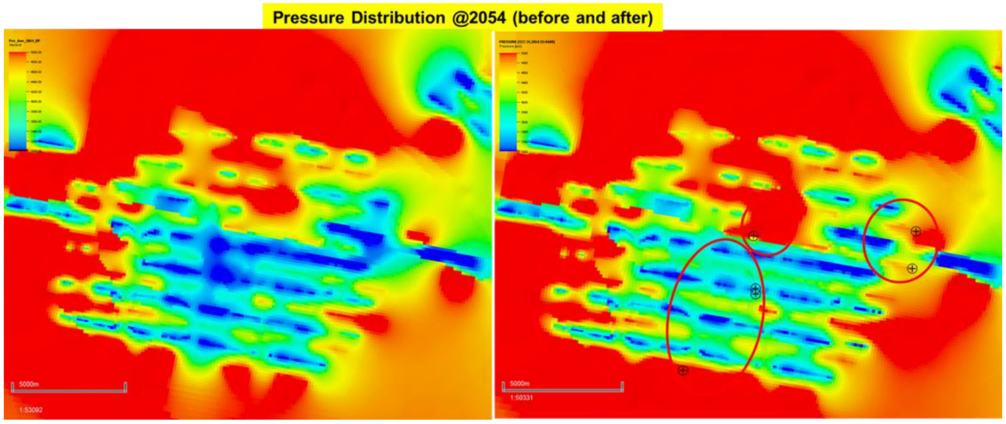

Figure 6—Enhancement in Reservoir Pressure After Well Pattern Optimization

Role of Digitalization and Artificial Intelligence Applications in Field Development

The digitalization and AI embracement program is projected to deliver an estimated $120 million in value creation, driven by a suite of cutting-edge technologies and innovative reservoir management solutions. Central to this transformation is the deployment of advanced EOR dashboards leveraging streamline simulation, which are updated monthly to calculate critical metrics such as allocation factors and pattern Voidage Replacement Ratios (VRR) as part of the IRM Smart Performance Review (SPR) process. These dashboards enable faster, data-driven decisions that optimize sweep efficiency and ensure consistent pressure maintenance across the reservoir.

Complementing this, a Reservoir Monitoring Plan Automation initiative has been launched, streamlining data collection, analysis, and reporting processes. This automation not only boosts operational efficiency but also ensures strict enforcement of IRM guidelines, reducing human error and enabling proactive interventions. Meanwhile, the Production and Injection System Optimization program introduces SRM standardization, harmonizing production practices across assets and refining VRR management to ensure optimal waterflood and gas injection performance.

A key breakthrough is Autonomous Control System (ACS), the world's first AI-driven Advanced Process Control (AI-APC) solution that achieves 100% reservoir guidelines compliance. ACS represents a paradigm shift in operating philosophy, moving from manual well adjustments performed by human operators to autonomous well control executed by artificial intelligence. This capability allows wells to self-adjust to near-optimum gas lift rates, unlocking an expected production gain of 1–2%while delivering a 5% reduction in gas lift consumption, translating to both incremental revenue and significant operational savings.

Further digital integration is achieved through real-time downhole monitoring, which maximizes the value of surveillance data by feeding live information into Artificial Intelligent Production System Optimization (AiPSO) models for continuous optimization. In parallel, proactive CO_2 monitoring is being implemented to enhance CO_2 sweep efficiency in EOR operations while minimizing the cost of CO_2 recirculation a critical balance for both performance and sustainability objectives.

Together, these initiatives not only represent a technological leap forward but also redefine how reservoirs are managed, shifting from reactive, manual operations to a data-driven, automated, and AI-powered environment. The cumulative effect is greater efficiency, reduced operating costs, and substantial production gains, underscoring how embracing digitalization and AI is transforming the business model and creating measurable value.

Fig. 7 shows examples of digitalization and artificial intelligence applications used for Reservoir-X development.

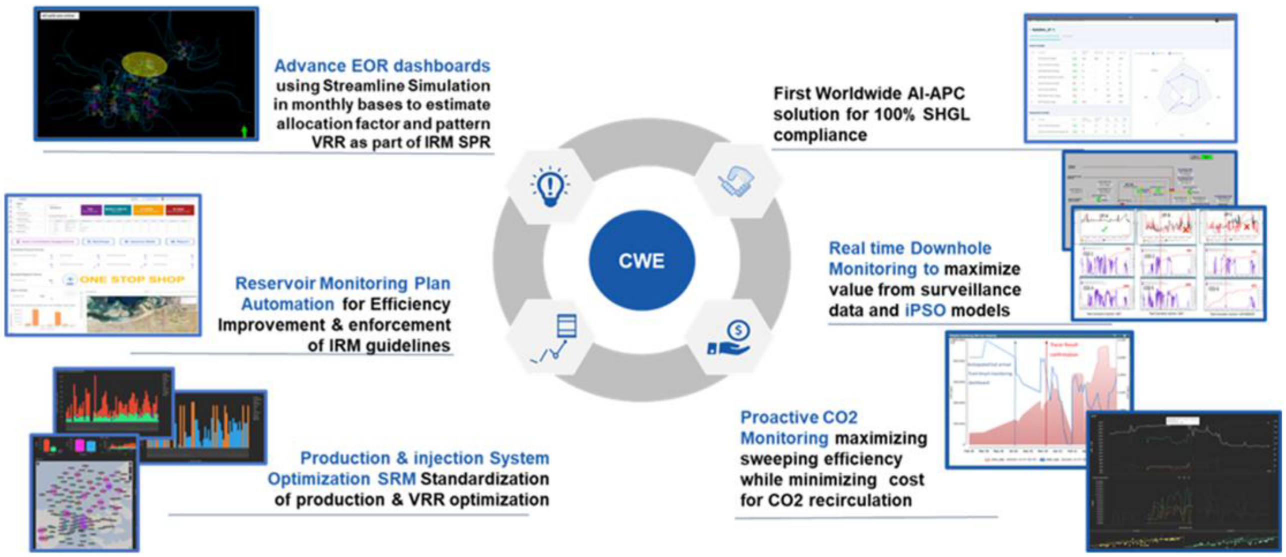

Figure 7—Examples of Digitalization and Artificial Intelligence Applications in Field Development

Conclusion

Field development reviews in mature reservoirs often reveal inefficiencies such as incomplete well patterns, poor sweep efficiency, and uneven pressure support, which can hinder recovery and long-term planning. To address these issues, a Well Pattern Optimization (WPO) approach was implemented, focusing on diagnosing pressure-depleted zones, improving injector-producer alignment, and leveraging digital technologies to strengthen reservoir management for Reservoir-X in one of the onshore oil fields in UAE.

The strategy combined reservoir monitoring, simulation analysis, and adaptive planning to enhance sweeping efficiency using existing infrastructure. Digital dashboards were introduced to track performance, identify deviations, and guide corrective actions, while an AI-driven automation system enabled wells to self-adjust operations improving control and responsiveness to changing field conditions.

The optimized pattern improved pressure distribution, reduced bypassed oil, and enhanced fluid displacement. Real-time surveillance enabled quicker interventions, while AI automation boosted operational efficiency and stabilized production performance.

This integrated approach combining pattern optimization, digital monitoring, and AI-enabled control demonstrates a scalable solution for mature fields. It highlights how aligning subsurface engineering with digital innovation can unlock additional recovery, extend field life, and build operational resilience.

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